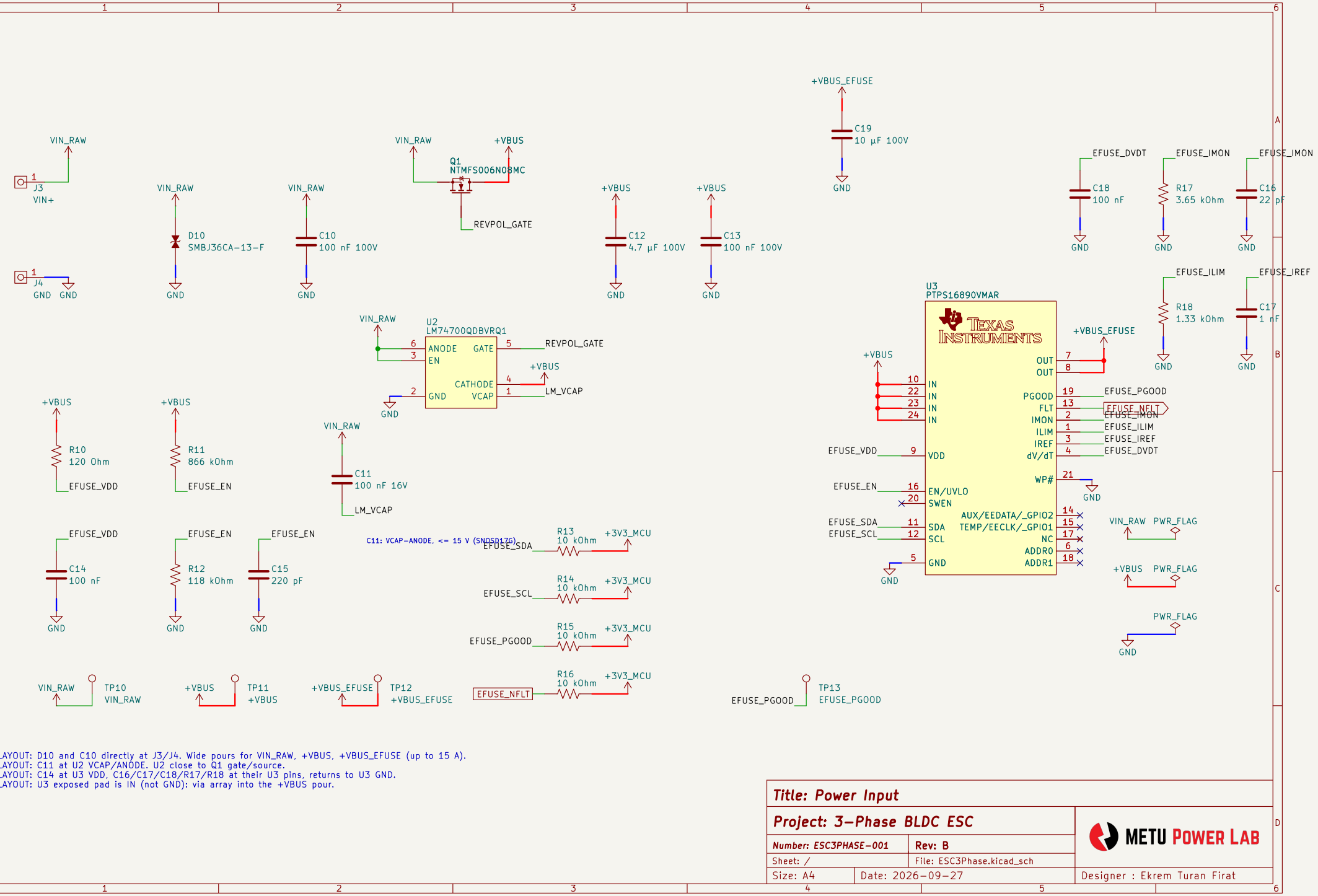

VIN_RAW
J3
VIN+
J4
GND
D10
SMBJ36CA-13-F
C10
100 nF 100V
Q1
NTMFS006N08MC
REVPOL_GATE
+VBUS
C12
4.7 µF 100V
C13
100 nF 100V
C19
10 µF 100V
+VBUS_EFUSE
C18
100 nF
EFUSE_DVDT
R17
3.65 kOhm
EFUSE_IMON
C16
22 pF
R18
1.33 kOhm
EFUSE_ILIM
C17
1 nF
EFUSE_IREF
U2
LM74700QDBVRQ1
ANODE
GATE
EN
CATHODE
VCAP
LM_VCAP
C11
100 nF 16V
U3
PTPS16890VMAR
TEXAS INSTRUMENTS
IN
OUT
PGOOD
FLT
IMON
ILIM
IREF
dV/dT
WP#
EN/UVLO
SWEN
VDD
SDA
SCL
AUX/EEDATA/_GPIO2
TEMP/EECLK/_GPIO1
NC
ADDR0
ADDR1
EFUSE_PGOOD
EFUSE_NFLT
EFUSE_VDD
EFUSE_EN
EFUSE_SDA
EFUSE_SCL
R10
120 Ohm
R11
866 kOhm
C14
100 nF
R12
118 kOhm
C15
220 pF
C11: VCAP-ANODE, <= 15 V (SNOSD176)
R13
10 kOhm
R14
10 kOhm
R15
10 kOhm
R16
10 kOhm
+3V3_MCU
PWR_FLAG
TP10
VIN_RAW
TP11
+VBUS
TP12
+VBUS_EFUSE
TP13
EFUSE_PGOOD
LAYOUT: D10 and C10 directly at J3/J4. Wide pours for VIN_RAW, +VBUS, +VBUS_EFUSE (up to 15 A).
LAYOUT: C11 at U2 VCAP/ANODE. U2 close to Q1 gate/source.
LAYOUT: C14 at U3 VDD, C16/C17/C18/R17/R18 at their U3 pins, returns to U3 GND.
LAYOUT: U3 exposed pad is IN (not GND): via array into the +VBUS pour.
Title: Power Input
Project: 3-Phase BLDC ESC
Number: ESC3PHASE-001
Rev: B
Sheet: /
File: ESC3Phase.kicad_sch
Size: A4
Date: 2026-09-27
METU POWER LAB
Designer : Ekrem Turan Firat